

Empirical Research Paper

Generative AI in project management: Impacts on corporate values, employee perceptions, and organizational practices

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ABSTRACT

This study examines the evolving role of generative AI tools, particularly ChatGPT, in project management, focusing on their impact on corporate values, employee perceptions, and practical application across project phases and roles. Using a mixed methods design comprising a literature review, two industry workshops, and a survey of 52 professionals from diverse sectors, the research integrates thematic qualitative analysis with exploratory quantitative assessment. The most prominent finding is that 74 % of participants expressed mixed or negative sentiments towards AI adoption, citing concerns about job security and data privacy, despite recognizing productivity and automation benefits. In addition, 42 % observed positive shifts in corporate values linked to AI adoption, and strong consensus emerged regarding AI's usefulness in Planning (86 %), Monitoring and Controlling (75 %), and Integration (83 %) project management process groups. While some statistically significant associations were identified, such as employer type (consultant vs. non-consultant) influencing AI use during the project execution phase, these findings are preliminary due to the small sample size. The results highlight the importance of balancing AI-driven efficiencies with ethical safeguards, human oversight, and targeted training. This study contributes to the digital transformation discourse by providing early empirical insights into how generative AI is reshaping project management practices and organisational culture.

1. Introduction

Generative Artificial Intelligence (AI) is sweeping through the project management landscape, heralded as a revolutionary tool for enhancing efficiency and decision-making, yet also feared as a “Pandora’s box” of job displacement, data insecurity, and ethical dilemmas (Simon et al., 2024; Dacre et al., 2025; Dwivedi et al., 2023). This dichotomy of opportunity and apprehension presents an unprecedented challenge for project leaders and organizations.

As AI technologies rapidly evolve, project-based industries are

grappling with how to integrate these tools without compromising human-centric values such as trust, collaboration, and ethical accountability (Bankins and Formosa, 2023). While much of the discourse around AI adoption has focused on technical feasibility and productivity gains, there remains a significant gap in understanding how project professionals themselves perceive and respond to these changes (Goldberg et al., 2023).

How are project management professionals, in a field that heavily relies on human judgment and collaboration, perceiving, adopting, and adapting to this disruptive technology? And what is the interplay

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between organizational-level shifts in values and individual-level employee sentiments? To conceptually ground this exploration, the study draws on the “Technology Acceptance Model (TAM)” (Davis, 1989), which proposes that technology adoption is shaped by two primary perceptions: usefulness and ease of use. These constructs are especially relevant for understanding how project professionals evaluate generative AI tools within the constraints and demands of real-world project environments. Despite the growing presence of generative AI in project workflows, empirical research on its acceptance and perceived value among project managers remains scarce (Felicetti et al., 2024). This study addresses that gap by combining qualitative and quantitative methods to capture both organizational and individual perspectives.

Through workshops conducted on generative AI with project practitioners, followed by a survey assessing the workshop participants’ perspectives, this study investigates the shifts in corporate values, changes in employee sentiments, and the perceived applicability of generative AI tools in project management. The goal of this study is to provide a valuable snapshot of how project management professionals are perceiving, using and reacting to a disruptive technology, to provide practical insights and guidance for effective integration of generative AI in project management practices. The insights gained from these workshops and the findings from a subsequent survey of 52 professionals offer a deeper understanding of the transformative potential of AI in project management, particularly in terms of organizational adoption, employee perceptions, and the evolving role of generative AI tools. This research contributes to the field by highlighting key learnings from the workshops, presenting empirical data on professional perceptions that can serve as a hypothesis-generating resource in the future, and offering practical recommendations for leveraging AI to enhance project management practices.

The remainder of the paper is structured as follows: Section 2 reviews relevant literature on AI adoption and project management; Section 3 outlines the research methodology; Section 4 presents the findings; Section 5 discusses implications; and Section 6 concludes with recommendations, research limitations, and future research directions.

2. Literature review

2.1. Overview

Recent advancements in AI and Natural Language Processing (NLP) have led to the development of sophisticated generative language models (Min et al., 2024; Ray, 2023; Han et al., 2021). The evolution began with early techniques like Hidden Markov Models (HMMs) and Gaussian Mixture Models (GMMs), followed by the introduction of Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks (Cao et al., 2023). A breakthrough came with the transformer architecture in 2017 (Vaswani et al., 2017), which enabled the development of powerful large language models (LLMs) such as BERT and GPT (Cao et al., 2023; Dasgupta et al., 2023). These models now underpin widely used tools like ChatGPT, Bard, and Gemini, delivering human-like text generation capabilities that are reshaping professional workflows across sectors (Mohamed et al., 2025a; Mohamed et al., 2025b; Gupta et al., 2024; Iorliam and Ingio, 2024; Roumeliotis and Tselikas, 2023).

Despite the growing interest in generative AI tools within project management, there is limited empirical research addressing their direct impact on project managers and employee productivity. A literature review was conducted to understand the existing body of knowledge and identify gaps in generative AI as applied to project management. This review was instrumental in shaping the development of workshops and surveys that are the methods of acquiring data for this paper. The literature review highlighted the potential of generative AI tools, such as ChatGPT, Google’s Bard, and Microsoft’s Bing Assistant, in various industries and project management, as well as challenges related to data quality and ethical considerations. Building on this foundation, the

integration of generative AI tools, particularly ChatGPT, into project management processes emerges as a needed area of inquiry (Weng, 2023). The subsequent sections explore the applicability, advantages, challenges, and opportunities of adopting generative AI in project management from the literature.

2.2. Applicability of generative AI tools in project management

Adopting generative AI tools, such as ChatGPT, Google’s Bard and Gemini, and Microsoft Bing Assistant into project management presents a broad spectrum of potential applications and advantages, transforming how organizations handle their projects (Weng, 2023). Aramali et al. (2024a,b,c) investigated the use of ChatGPT across the 10 Earned Value Management System (EVMS) subprocesses which are based on the National Defense Industrial Association (NDIA) EVMS guidelines (NDIA, 2018); the study reveals that ChatGPT can enhance compliance and efficiency in these subprocesses, which align with the maturity and environment project controls frameworks (Aramali et al., 2024a,b,c) by improving data analysis and providing automated solutions to streamline project controls. On the other hand, the Project Management Body of Knowledge (PMBOK) Guide organizes project management processes into five distinct project stages called “Process Groups”: Initiating, Planning, Executing, Monitoring and Controlling, and Closing (Project Management Institute, 2021). Integrating generative AI in project stages can redefine stakeholder roles and automate processes, allowing project managers to focus more on strategic decision-making and innovation (Rane, 2023).

For the project initiation phase, generative AI is utilised to automate project requirements identification, goals setting, and scope definition, thus ensuring a solid foundation for project execution (Ma Chuan, 2023). Generative AI also showcases versatility by aiding in drafting project charters that effectively outline project objectives and scope (Weng, 2023). Barcaui and Monat (2023) compare the performance of the generative AI model “ChatGPT” with that of “Human Project Managers” in formulating project plans and schedules; they highlight ChatGPT’s ability to improve quality planning, showing how it complements the inputs of human managers through skilled prompt engineering. In the execution stage, ChatGPT facilitates the management of project activities by automating task assignments based on real-time project progress and resource availability, ensuring that project tasks are completed within the timeframe and budget (Abbas et al., 2023). Capabilities such as real-time data analysis, predictive and prescriptive analytics, and automated reporting enable ChatGPT to improve monitoring processes, hence maintaining project integrity and timelines (Ali Mohamed et al., 2023). Other assistance from generative AI models can be applied to project closing, post-project reviews, releasing resources, and enabling the systematic collection of lessons learned for future project improvements (Weng, 2023).

In addition, generative AI can improve project controls supporting the application of NDIA guidelines in integrated project/program management (NDIA, 2018). Specifically, the Integrated Project/Program Management (IP2M) Maturity and Environment Total Risk Rating (METRR) tool is developed by the US Department of Energy (DOE) to make EVMS more reliable. The goal was to develop an automated assessment tool that could be used for their internal projects and resolve project issues. Subsequently, Aramali et al. (2023) presented in a few industry workshops the performance and accuracy of project control practices of using OpenAI’s ChatGPT and Google Bard. The feedback from the workshops indicated that ChatGPT has significant potential to enhance risk management, generate variance analysis narratives, and find policies and guidelines in integrated project management. Its effectiveness depends on data quality and user instructions, with concerns about deep analysis and identifying root causes (Aramali et al., 2024a,b,c). However, successful adoption hinges on addressing challenges related to data privacy, algorithmic bias, and other implementation barriers (Ayinde et al., 2023). The future of project

management will see a more integrated utilization of generative AI tools, which necessitates further research and development to fully understand their potential enhancements and breakthroughs as well as associated risks and challenges (Rane, 2023).

2.3. Challenges and opportunities of adopting generative AI

Many early adopters want to use generative AI to enhance their work efficiency. Seo and Kang (2024) analyzed the performance of a few NLP algorithms for automatically summarizing construction dispute precedents. They found that the performance of four models ranged from about 28 % to 36 %. Despite the modest performance, these results demonstrate the potential for NLP models to assist in the labour-intensive process of document summarization, suggesting a promising future with further advancements. Similarly, Goyal et al. (2023) assessed GPT-3 and other NLP models to summarize datasets from CNN, Daily Mail, BBC, and Newsroom. They found that the overall quality of summarized output, based on various performance metrics from the literature such as quality and factuality, ranged from 50 % to 60 %, indicating significant room for improvement. While some may argue that current generative AI models do not consistently provide high-quality summaries, the continuous advancements in this field hold promise for future enhancements. For example, H. Kim et al. (2021) suggest document management practices to improve classification accuracy for construction disaster data. These improvements are expected to significantly benefit project management tasks such as report generation, summarizing meetings, and project documentation, making these processes more efficient and accurate over time.

Based on the literature, generative AI offers substantial technical opportunities such as enhanced decision-making, as it can analyze large datasets to identify patterns, trends, and potential risks (Rane, 2023). In the rapidly changing environment of project management, the rising popularity of generative AI technology promises disruptive potential that might reshape existing paradigms. Fui-Hoon Nah et al. (2023) investigate this technological frontier, examining the double-edged sword of opportunities and challenges that adopting generative AI tools in project management processes presents, as it can revitalize activities such as handling sensitive data of organizations, thus leading to new data privacy and data security concerns (Ayinde et al., 2023). With generative AI, the risk of cyberattacks and data breaches also increases due to the new data structure and data interchange requirements (Ma Chuan, 2023). Data quality and data availability are other concerns when aiming to integrate generative AI in the project management process, as the generative AI models require extensive high-quality data to be trained. In complex projects, data is often incomplete and unstructured (Auth et al., 2021).

Rane (2023) also outlines ethical issues like algorithmic biases, leading to inappropriate responses and biased decision-making. Ethical and legal issues related to transparency in how data is collected, used, and shared with AI systems, and whether the decisions or responses could be audited or justified to the project stakeholders, are challenging while implementing generative AI in the sensitive domain of project management (Ayinde et al., 2023). Another ethical challenge in project management is the potential over-reliance on AI tools, which may undervalue or overlook human judgment and expertise as organizations increasingly rely on AI for their decision-making (Weng, 2023). All these challenges and opportunities call for a human-centric approach that also addresses ethical concerns and data security.

2.4. Employee's perspective towards integrating generative AI in project management

Project managers acknowledge the inevitable integration of generative AI in project management and perceive ChatGPT and similar generative AI tools as enhancing and enabling technical tools rather than replacements for their roles. Nguyen and Scheff (2023) highlight a

general optimism among project managers regarding the integration of AI into their professional workflow. Despite the optimism, some early adopters of AI express concerns about AI's ethical implications and the potential risks of replacing certain job functions (Cardon et al., 2023).

There exists a demarcation between the tasks that ChatGPT can automate and those that still require human capabilities, such as trust-building, empathy, and understanding team dynamics and stakeholder needs. However, there is also an underlying anxiety about AI's potential to replace human roles in the field of project management (Nguyen and Scheff, 2023). These job security concerns call for strategies to make the employees consider generative AI tools as partners in their work, rather than competitors (Joskowicz and Slomovitz, 2023). There exists a clear literature gap in how AI is perceived by employees and end-users and how it is used in project management and various other fields of application (Al Naqbi et al., 2024). To address this gap, investigating employee engagement with generative AI tools, as well as understanding their capabilities and challenges, is crucial for leveraging their potential effectively within the workplace (Ooi et al., 2023). Employee sentiments towards using generative AI in project management vary widely, from enthusiasm to scepticism, influenced by their familiarity with technology and concerns about AI's impact on their roles and responsibilities (Fui-Hoon Nah et al., 2023). According to Hernández Noé Chávez et al. (2024), to improve collaborative dynamics between the employees and generative AI tools, organizations must improve their processes, infrastructure, and resources to keep up with technological advancements and promote an efficient and ethically responsible work environment.

The theory of TAM has been widely applied to explain user behaviour and attitudes toward adopting new technologies, particularly in workplace settings. For example, Gefen and Straub (1997) used TAM to explore how employees perceived and adopted email systems, revealing that perceived usefulness and ease of use significantly influenced adoption behaviour. Similarly, Gefen et al. (2003) extended TAM to include the trust factor in online shopping environments, demonstrating how contextual variables can shape technology acceptance in professional settings. Perceived usefulness and perceived ease of use are central constructs in understanding employee receptiveness to tools like generative AI (Venkatesh and Davis, 2000). In the context of project management, these constructs can help interpret the diverse sentiments of project professionals, ranging from enthusiasm about efficiency gains to concerns over usability, control, and job displacement. TAM provided a useful conceptual framework to situate the observed variation in employee perceptions reported in this study.

2.5. Literature review findings and gaps

The review of the literature highlighted the potential of generative AI in enhancing project management practices mainly to support project decision-making processes. It also underscored the potential benefits of utilizing generative AI in managing project activities, such as empowering project managers to maintain project integrity and timelines more effectively. However, literature gaps still exist in understanding organisational dynamics in response to adoption of generative AI in the workplace. These gaps are particularly evident in three areas that informed the design of the questionnaire used in this study.

First, previous studies have shown that AI adoption can influence organisational ethics, transparency, and strategic priorities, which guided the development of survey items measuring perceived changes in corporate values following the introduction of generative AI tools (Ayinde et al., 2023; Fui-Hoon Nah et al., 2023).

Second, the Technology Acceptance Model, together with research on employee perceptions of AI, has identified perceived usefulness, ease of use, trust, and job security as important determinants of employee attitudes, which informed questions exploring enthusiasm, concerns, and confidence regarding the use of generative AI in project management (Gefen and Straub, 1997; Venkatesh and Davis, 2000; Nguyen and Scheff, 2023; Cardon et al., 2023).

Third, literature discussing AI-driven changes in project processes and governance provided the basis for survey items examining organisational practices, including operational adaptations, process automation, and the implementation of AI governance measures (Weng, 2023; Aramali et al., 2024a,b,c; NDIA, 2018). This explicit connection between the literature and the questionnaire ensured that the survey was grounded in established findings while directly addressing the identified gaps related to corporate values, employee sentiment, and organisational practices in the context of generative AI adoption in project management.

3. Research questions and methodology

Guided by the TAM framework, the following research questions were developed to explore how perceived organizational shifts and employee perceptions align with the broader dynamics of generative AI adoption in project management.

- (1) To what extent do changes in corporate values and strategic approach, driven by the evolution of generative AI technologies, influence employee sentiments regarding generative AI adoption in project management?
- (2) How do these organizational shifts impact the perceived applicability of generative AI tools in specific project management processes and roles?

The purpose is to provide further insights into the relationship between evolving organizational dynamics, employee perceptions, and the integration of generative AI technologies within the realm of project management. Given the early stage of generative AI integration in project management, this study takes an exploratory approach, using descriptive analyses to identify patterns and perspectives that can guide future research and organizational decision-making. The main research steps undertaken in this investigation are provided in Fig. 1.

3.1. Literature review and problem identification

In a preliminary study, the authors conducted an exploration of

generative AI in early 2023, a time when these tools were emerging and gaining popularity (Aramali et al., 2024a,b,c). The preliminary study focused on the application of generative AI in Integrated Project Management (IPM), particularly in the context of earned value management systems (Aramali et al., 2021). A few scholars have studied AI or machine learning techniques utilization within earned value management processes to more accurately predict changes in project schedule and budget during project execution (Balali et al., 2020; Batselier and Vanhoucke, 2017). However, the exploration and application of these technologies in IPM, particularly in the context of EVMS, remained limited. This initial phase involved identifying gaps in the literature focusing on the research direction on IPM, and consulting industry professionals to refine the scope. The preliminary study involved early discussions regarding generative AI with industry professionals, aimed to understand potential applications, limitations, and implications in the realm of IPM. To support this, the authors conducted a pilot workshop with 25 professionals, who engaged in interactive discussions on “ChatGPT and AI in Project Controls.” This workshop helped shape the future research questions and validated the relevance of the topic. They disseminated their early findings in a conference proceeding (Aramali et al., 2024). As part of the current study, the authors expanded the literature review beyond the narrow scope of earned value management systems. Building upon this initial phase, the authors had an opportunity to explore generative AI applications across broader project management processes and process groups as described in the PMI framework (Project Management Institute, 2021).

The current study delves into employee sentiments regarding new changes in the industry and explores various industry roles that could be impacted by these emerging technological advancements. The authors continued to conduct a comprehensive literature review on the broader project management processes, building upon the results of their preliminary study. The early 2023 literature lacked extensive sources on generative AIs across diverse fields, especially within the domain of integrated project management. Consequently, the research questions were crafted, aligned with the latest publications, and the identification of existing literature gaps. Additionally, the TAM provided a theoretical foundation for shaping the research questions, particularly in examining how perceived usefulness and ease of use influence project

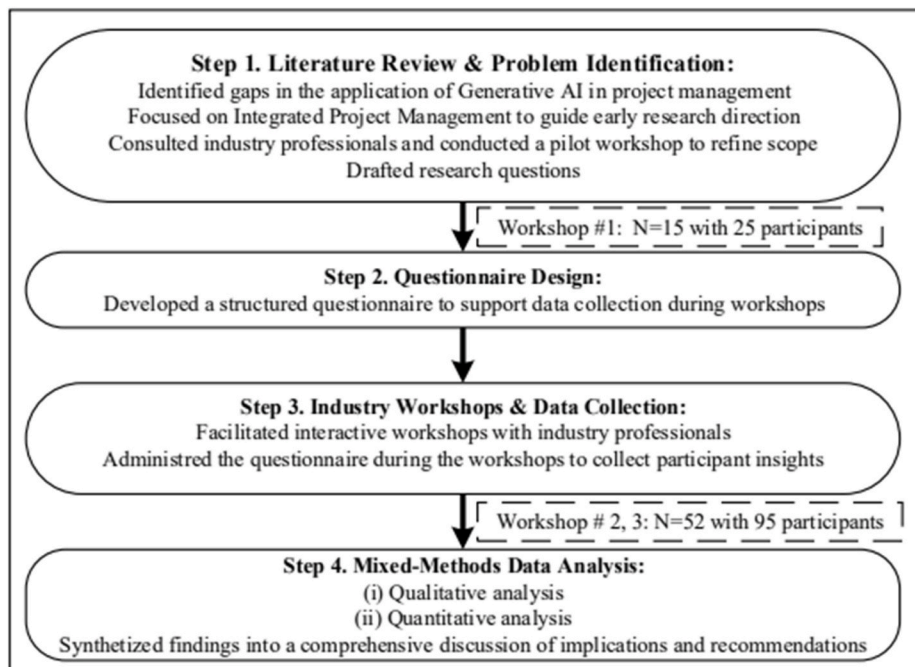


Fig. 1. Adopted research steps.

professionals' attitudes toward adopting generative AI tools. The findings from this literature review were discussed in the previous sections, shedding light on the current state of knowledge and theoretical underpinning and pinpointing areas where further exploration is demanded.

3.2. Questionnaire design

To comprehensively explore existing aspects around the generative AI tools in project management, the authors developed a questionnaire that was informed by the literature review and preliminary study results. It addressed existing research gaps identified by insights from expert discussions across multiple workshops. This iterative process combined scholarly understanding with practical industry insights, enhancing the questionnaire's effectiveness (Yardley et al., 2016).

The survey questionnaire, of which an overview is shown in Fig. 2, was developed based on previous research about IPM, with the main body of questions revised to focus on generative AI (Aramali et al., 2021). The first section collects demographic information such as participant's position, employer type, firm size, employment role, and sector(s) that are relevant to project management practice. Aligned with the research questions, the questionnaire checks participants' familiarity with generative AI tools and their perspectives on organizational shifts in corporate values due to AI advancements (Budhwar et al., 2023). Questions regarding employee sentiments on such tools provide a qualitative understanding of the human aspect of AI adoption (Fui-Hoon Nah et al., 2023). Then, the survey delves into specific phases and roles within project management processes (Project Management Institute, 2021) where generative AI or ChatGPT could be applied, by prompting

participants to envision a few scenarios and draw from their personal experiences.

The survey touches upon human-AI interaction (the collaboration dynamics between AI-powered tools and human team members), strategies for smooth AI adoption, and participants' foresight on the role of generative AI tools in project management (Weng, 2023). The survey concludes with participants sharing their final thoughts on potential AI advantages and challenges, allowing for general comments.

3.3. Industry workshops and data collection

Two primary workshops were organized virtually in 2023, attracting 95 participants from the project management community with a keen interest in contemporary developments in generative AI. These workshops served as a platform to foster community engagement on pertinent subjects and facilitated data collection through voluntary survey participation. To integrate qualitative and quantitative insights, the survey was administered to attendees during the workshops. Throughout the workshops, participants willingly and confidentially responded to questionnaires, yielding the collection of 55 new datasets. After data screening, $N = 52$ datasets included complete responses.

3.4. Mixed-methods data analysis

Analytical approaches were employed to analyze the survey data, tailored to the nature of the survey questions on demographics, closed-ended Likert-scale queries, or open-ended inquiries. Demographic questions aimed to enhance data credibility and facilitate internal comparisons. Quantitative analysis was conducted using Excel™ and

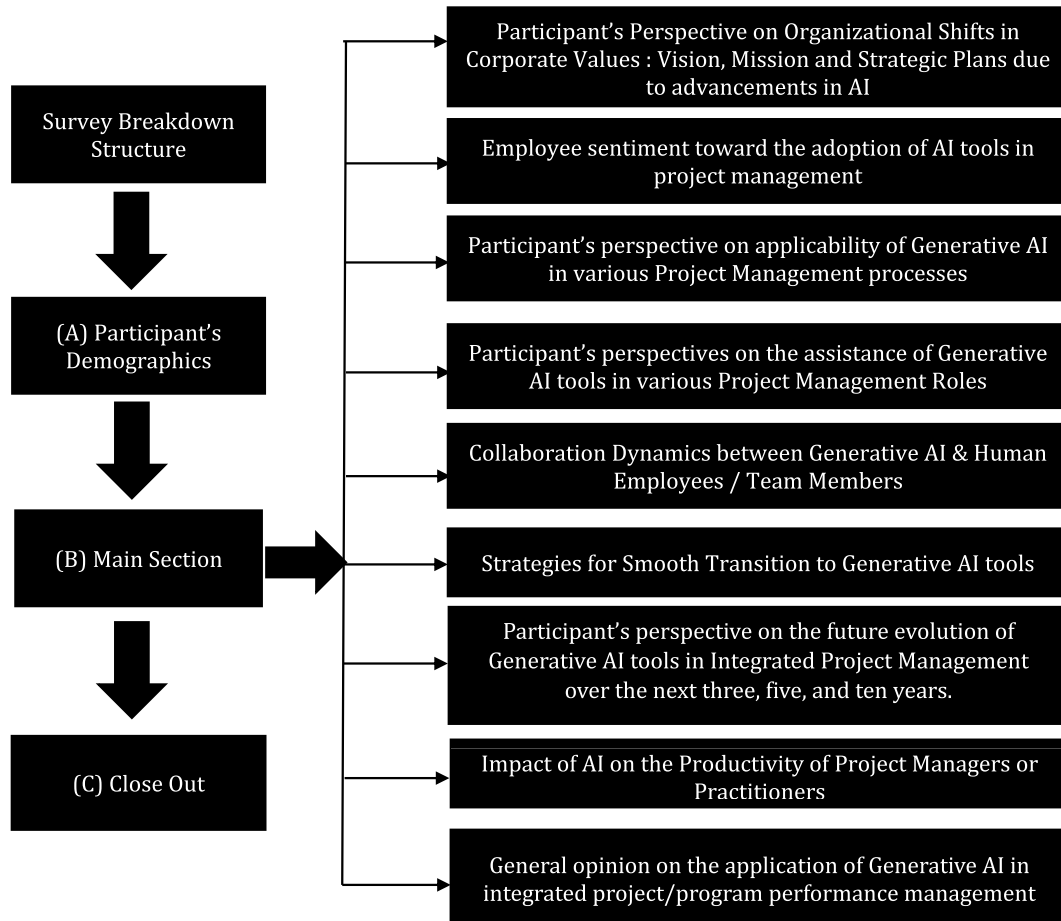


Fig. 2. AI adoption in project management; questionnaire overview.

Python (Haslwanter, 2022), while qualitative analysis employed Nvivo software (Zamawe, 2015). The quantitative analysis included descriptive analysis, including frequencies and cross-tabulations, and statistical analyses. The qualitative analysis involved coding, querying the most repeated keywords, and automated coding to validate the insights (AlYahmady and Al Abri, 2013). In summary, the data were analyzed thematically in addition to the quantitative approach. As this study involved a small sample size, the role of the quantitative analysis was strictly exploratory and descriptive, aimed at illustrating potential patterns rather than testing formal hypotheses or establishing causal inferences.

In the quantitative analysis, the demographic variables that included the organization type (public vs. private), employer type (consultant vs. non-consultant), years of experience in project management (greater than 25 years or lower), whether AI impacted corporate values or not, employee sentiment on AI (positive or negative), whether the professional is experienced in one sector (domain) or multiple, the size of firm (greater than 500 employees vs lower), different employment positions in the field, and familiarity with ChatGPT or generative AI tools (yes or no) were considered as independent variables. These variables align with their established significance in project management practice, substantiated by consensus among 36 industry experts in a previous research by the authors (Aramali et al., 2024a,b,c).

The study's dependent variables included the ten project management knowledge areas and five project management process groups as defined by the Project Management Institute (2021). Specifically, participants provided assessments on a 5-point Likert scale for the applicability of ChatGPT in various project management phases, as well as the extent to which generative AI tools could assist different project management roles (Joshi et al., 2015). The Likert scale for closed-ended questions on the impact of generative AI tools on project management processes ranged from "Definitely not" to "Definitely yes" for specific phases, and from "Strongly disagree" to "Strongly agree" for different positions and roles, allowing participants to express their perspectives on the potential utility of AI tools in these contexts.

In the analysis phase, the responses were categorized into two groups: "high," comprising those who selected "Strongly agree," "Agree," "Definitely yes," or "Probably yes," and "low" for those who chose otherwise. For comprehensively understanding the perceived applicability of AI in project management process groups and knowledge areas, the internal consistency of the surveyed items was checked using Cronbach's alpha. The obtained Cronbach's alpha of 0.92 underscored the reliability and internal consistency of the items (Tavakol and Dennick, 2011).

Roles such as project controllers, project schedulers, project planners, project control account managers, project managers, consultants, researchers, and project management educators/trainers also went through a similar analysis. Then, in order to explore potential relationships between each demographic variable and the dependent variables, Fisher's exact test was applied, allowing for a comprehensive examination of their associations iteratively (H.-Y. Kim, 2017). These tests were conducted to identify potential signals or emerging trends in the data, and the results are presented as preliminary and illustrative rather than confirmatory.

4. Results and findings

4.1. Characteristics and perceptions of respondents

The 52 participants in this study came from a diverse array of positions, with project managers constituting the largest group (11 participants), followed by project schedulers (3), chief executive officers (3), and program managers (3), among others. In terms of employer types, many participants identified themselves as consultants (23), followed by government contractors and employees (19), industry contractors (6), and manufacturers (4). The sizes of firms for which the participants

worked for varied, with 32 participants employed in large organizations having over 500 employees, while 20 participants worked in relatively smaller firms. The participants spanned over multiple unique sectors, with Information Technology (IT) being most dominant (13), followed by Aerospace (5), Construction (4), Defense (3), Science (3), and various combinations of these sectors.

In evaluating the impact of AI on corporate values, 58 % of participants noted no significant shifts, while 42 % reported positive changes in response to a binary (Yes/No) question asking whether their firm's vision, mission, or strategic plan had shifted due to AI advancements. Regarding employee perceptions of AI application in project management, a majority (74 %) expressed mixed or negative feelings, with only 26 % reporting positive sentiments. These participants represented diverse levels of industry experience, with an average of 17.12 years. When considering the distinction between high and low experience levels based on a 25 years cutoff (Aramali et al., 2021), 58 % of participants fall into the high experience category, while 42 % fall into the low experience category. Most participants, 65 %, were from private organizations, while the remaining 35 % represented public entities. Furthermore, concerning ChatGPT or an analogous generative AI tool utilization, we found that 73 % of participants had actively employed these tools by the time we collected survey results, while the remaining 27 % were aware of them and expressed interest in utilizing them. This participant profile captures various dimensions of professional backgrounds within the project management and AI landscape.

Fig. 3 presents the resulting outcomes concerning participants' perceptions regarding the suitability of ChatGPT or other similar generative AI tools across different project management processes and knowledge areas. The stacked-bar chart in this figure illustrates the percentages of practitioners who agree on AI applicability per specific project management process groups or knowledge area, versus those who did not agree. In the initiation process group, 67 % agreed on the applicability of AI tools, while the percentages observed for subsequent process groups were 86 % for planning, 59 % for execution, 75 % for monitoring and controlling, and lastly, 59 % for closing process groups. These percentages indicate a consensus among participants regarding the potential usefulness of AI tools across different project management stages. This widespread agreement suggests that integrating AI tools into project management processes could enhance efficiency and effectiveness. Moreover, when assessing specific areas such as project integration, scope, schedule, cost, quality, resource, communications, risk, procurement, and stakeholder management, most participants consistently favoured AI assistance, with percentages ranging from 66 % to 83 %. Similarly, participants expressed their views on the extent to which generative AI tools could assist various project management positions (Fig. 4).

Overall, there was a positive inclination towards AI assistance, with percentages ranging from 51 % to 80 %. Particularly, project managers (80 %) and project management educators/trainers (80 %) expressed the highest endorsement for AI assistance, suggesting a strong belief in the potential benefits of AI in these roles. Lastly, when asked about the anticipated impact of AI on productivity, 52 % of participants believed that AI could lead to productivity increases beyond 20 %, while 47 % were more sceptical, foreseeing marginal to no productivity enhancements. This division reflects a spectrum of optimism levels among participants regarding the transformative potential of AI in project management practices.

Fig. 5 presents survey data on professionals' perceptions of AI in the workplace, specifically regarding collaboration dynamics and productivity. The first chart (on the right) indicates a balanced view towards AI collaboration, with a preference for a moderate human-AI task distribution (25 %) and a notable percentage of respondents (23 %) undecided about AI's role. The second chart (on the left) echoes this moderate stance, with the largest group (30 %) expect AI to contribute up to a 40 % increase in productivity. Overall, both charts reveal an optimism but also a cautious attitude towards generative AI utilization in project

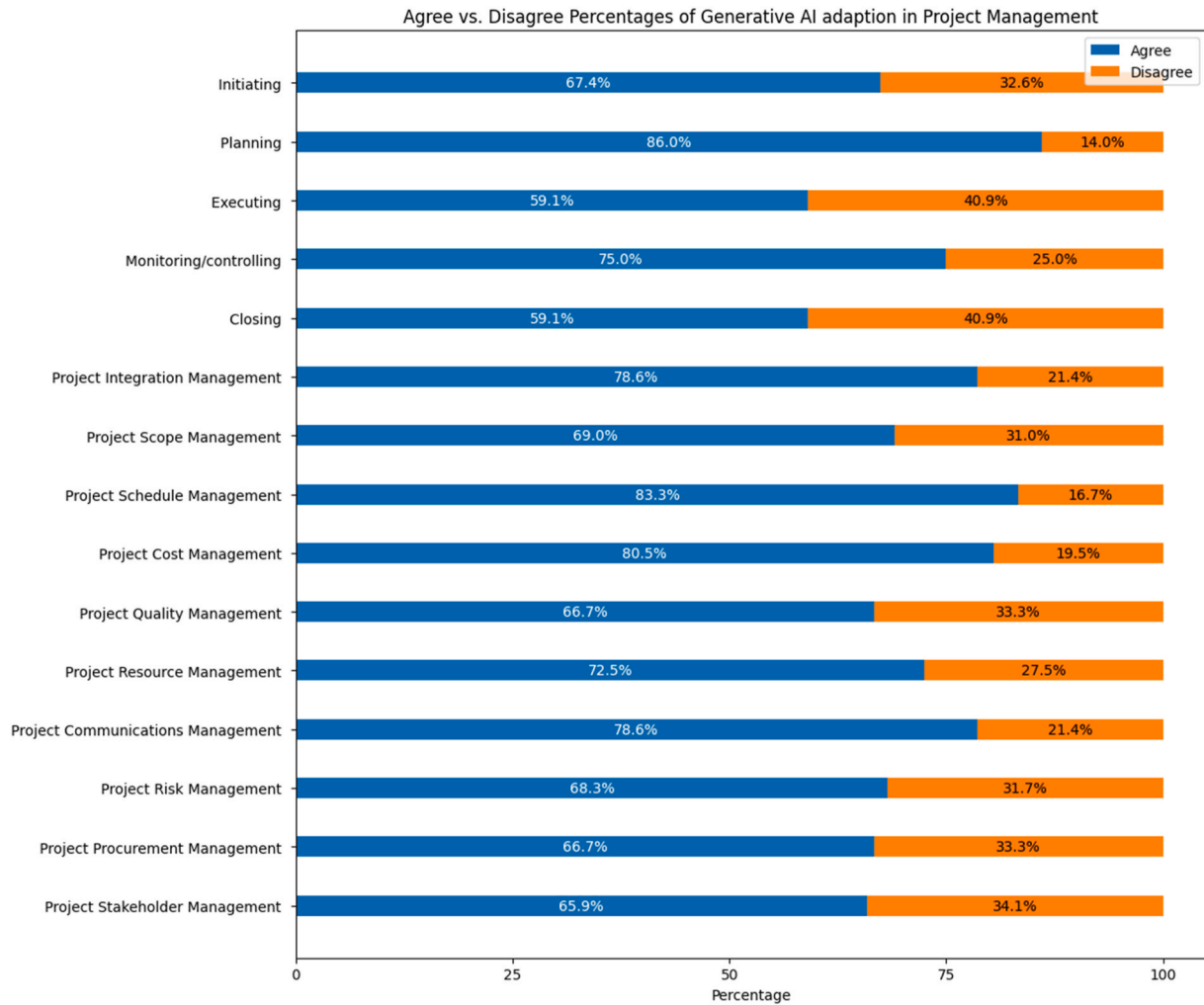


Fig. 3. AI adoption in project management.

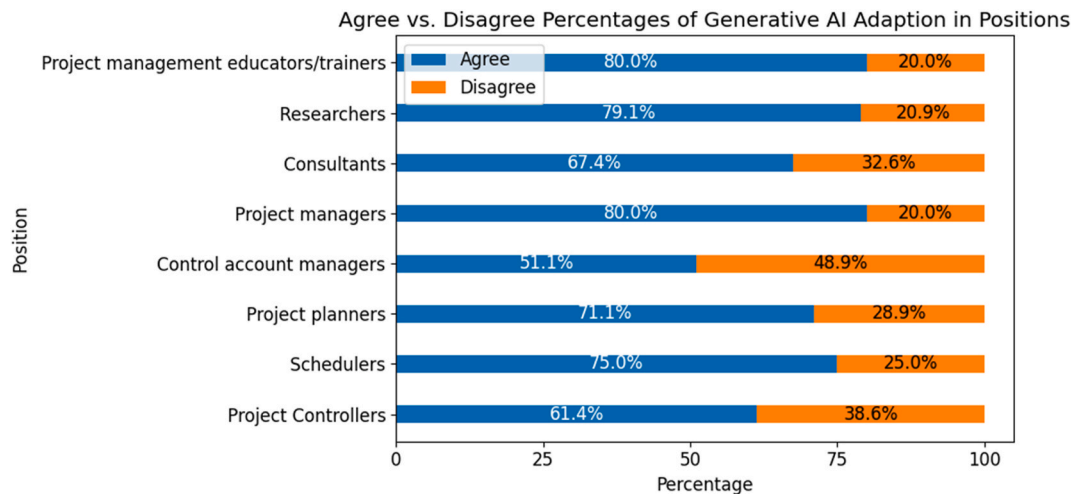


Fig. 4. Generative AI adaption in project management positions.

management processes, recognizing its potential to enhance workplace efficiency while also acknowledging the current limitations and uncertainties surrounding its integration.

4.2. Correlations

The correlations between demographic variables and the different project management process groups and knowledge areas, as well as different categories of roles, were examined through 184 Fisher's exact

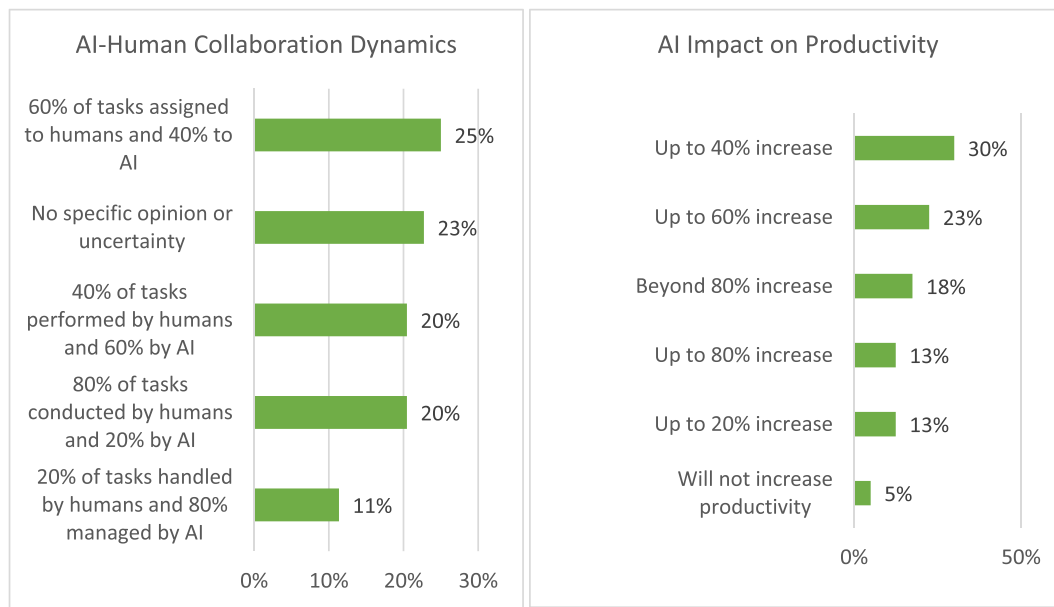


Fig. 5. Generative AI vs. humans.

statistical tests. The iterative approach of the analysis aimed to uncover significant associations, with p -values serving as indicators of statistical significance. To facilitate identification of the most compelling correlations, the p -values were mapped onto a dot plot, allowing for the quick discernment of the strongest associations. The results identified very few significant correlations where p -values were less than 0.1 and the significance threshold of 0.05 as reported in this section. Also, given the large number of statistical tests conducted relative to the small sample size, these findings are intended to be exploratory in nature.

While a few associations reached conventional thresholds of significance, the potential for Type I errors due to multiple comparisons cannot be ruled out. These results should be interpreted with caution and viewed as hypothesis-generating rather than confirmatory. Future studies with larger and more representative samples are needed to validate and extend these initial observations. To enhance accessibility of the findings, we have included a summary table (Table 1) presenting the most statistically significant correlations (p -value < 0.1), along with their corresponding odds ratios. This table highlights the variables with the strongest observed relationships and serves to complement the more detailed narrative discussion that follows.

The significance of these findings lies in identifying where AI assistance is currently being integrated into project management process groups, knowledge areas, and roles compared to one another and in

revealing the demographic and organizational factors that can influence this integration. Specifically, the analysis examines various factors influencing the different project management phases (i.e., process groups), including Initiation, Planning, Execution, Monitoring and Controlling, and Closing, as well as specific knowledge areas such as integration, scope, schedule, cost, quality, risk, resource, procurement, communications, and stakeholder management.

Several variables, including organization type, employer type, sector, years of experience, prior usage or familiarity with ChatGPT or another generative AI tool, perception on AI impact on corporate values, and participant's sentiment towards AI in project management, were examined for their association with each project management phase and aspect; particularly, analysing the odds ratios and p -values across the variables and project management phases. The results demonstrated that employer type significantly influences the execution phase (p -value = 0.04, which indicates a strong association), whereas it has less influence on other phases. This implies that whether an individual works as a consultant or non-consultant can affect the suitability of generative AI tools particularly during the execution phase of project management processes. Moreover, consultants tend to favour the utilization of these tools specifically during project execution. On the other hand, AI impact on corporate values presents a significant association with project integration (p -value = 0.004) and project quality management knowledge areas (p -value = 0.05), implying that shifts in corporate values due to AI significantly impact these management aspects. Specifically, companies acknowledging shifts in their values due to AI technologies and emerging trends are more inclined to integrate AI solutions to enhance project integration and improve project quality management.

Additionally, years of experience significantly influences the applicability of generative AI tools in project planning, scope management, and schedule management phases, as indicated by p -values less than 0.05 which suggests a strong association. The endorsement pattern suggests that individuals with less experience in project management more strongly endorse using these tools for project planning, scope management, and schedule optimization. With respect to studying influences on the project management roles, first, a significant association was found between firm size and the integration of AI into the project manager role (p -value = 0.0072). This suggests that larger organizations might be more inclined or better equipped to adopt AI technologies among project managers. Second, a significant correlation was found between individuals using ChatGPT and their inclination as consultants

Table 1
Summary of significant correlations.

Independent Variable	Dependent Variable	p-value	Odds Ratio
ChatGPT Familiarity (0 = No, 1 = Yes)	Initiating Phase	0.08	5.00
Years of Experience	Planning Phase	0.01	15.63
Employer Type	Execution Phase	0.04	4.67
ChatGPT Familiarity	Integration Phase	0.06	5.20
AI Impact on Corporate Values	Integration Phase	0.004	∞
AI Impact on Corporate Values	Quality Management	0.05	4.55
Years of Experience	Scope Management	0.007	8.8
Years of Experience	Schedule Management	0.03	7.5
Firm Size	Project Manager Role	0.0072	14.55
ChatGPT Familiarity	Consultant Role	0.003	6.86
ChatGPT Familiarity	Research Role	0.05	5.4

or researchers to seek AI assistance integration (p-values of 0.003 and 0.05, respectively). This suggests that familiarity with generative AI tools like ChatGPT increases acceptance of AI assistance adoption among consultants and researchers.

Overall, these findings underscore the importance of considering diverse factors and their interplay in various project management phases and adoption of AI for effective project execution and success. However, it is crucial to remember that the presence of a statistically significant p-value does not confirm causality. These results imply potential relationships that merit further investigation to understand the dynamics and causal directions of these associations. The study's extensive nature, covering a broad range of factors, provides a rich dataset for such future explorations.

5. Discussion

The survey participants highlighted various insights and considerations that they perceive as important for leveraging generative AI tools in project management. These insights have been thematically summarized into four distinct broad areas: automation; need for planning and training; employee sentiments; and finally, improved employee productivity. Other thoughts were also shared by respondents which are summarized in the last subsection. The subsequent paragraphs encapsulate the resulting outcomes derived from the qualitative analysis conducted with software and iteratively validated by the authors, as detailed in the methodology section. To support transparency and interpretive depth, anonymized quotes from participants were also included to illustrate the key themes that emerged from the qualitative data.

5.1. Automation

The theme of automating routine tasks to streamline operations and enhance efficiency permeates the insights gathered from the responses, highlighting a significant trend toward embracing generative AI and digital transformation. As one participant explained, *"Project management is inherently very inefficient without AI and automation. Lots of areas where repetitive tasks can be automated and AI can be used to target critical aspects of a project like risk management, quality management and scheduling"*. This sentiment was echoed by another, who observed, *"I think ultimately many tasks and interactions that PMs will have with stakeholders will be automated, resulting in PMs overseeing to ensure things stay on track and ensuring that the Chatbots are aligned with the Project goals"*. The pursuit of automating routine tasks is primarily driven by the desire to save time and money (i.e., improving project schedule and project budget), underscoring the efficiency and optimization AI tools can bring in operational processes. Beyond simple task automation, these tools can reshape job descriptions and functions. The technology's ability to perform tasks traditionally handled by staff, such as emailing and other administrative duties, showcases its impact on reducing the need for certain positions, thus altering workforce requirements. The findings by a recent study by Rane (2023) on task automation is aligned with these insights. Moreover, generative AI tools can support tasks where explanations of complex charts, metrics, and narratives are necessary with unprecedented speed and accuracy.

Current research supports this finding more solid. For example, Zheng et al. (2023) use generative AI to interpret complex clauses in building codes and Zhou et al. (2023) developed a model to find construction project contract missing clauses. This transformation enables more informed decision-making processes. When it comes to organizations restructuring to incorporate digital transformation and governance roles explicitly designed to leverage AI and generative technologies, it not only changes the way businesses operate but also enhances customer engagement: firms can collect field data, predict product failures more precisely, and refine their services. There is an increased interest in a broader commitment to leveraging AI's capabilities, understanding

them more deeply, with specific discussions on them for government purposes. Furthermore, the responses highlighted that the industry's interest is in the ongoing innovations in AI, announcing software updates that integrate AI for enhanced project management (e.g., variance analysis) and decision-making processes. These insights collectively underscore the theme of automating routine tasks by leveraging generative AI as a critical driver of operational efficiency and customer engagement strategies, backed up also by only recent studies (e.g., Allioui and Mourdi, 2023).

5.2. Need for planning and training

The collected insights reveal a challenge in the widespread adoption and integration of AI technologies within organizations, highlighting a clear theme: the critical need for planning and training. Several key issues characterize this challenge. First, there's a lack of strategic planning; many organizations recognize the potential of AI but have not implemented or developed strategies to harness its capabilities and thus, fail to acknowledge its full implications. This limited recognition is coupled with insufficient policy and guidelines. One participant emphasized this gap, stating, *"Lot of companies limit or restrict how we use these tools. In my company most of these tools are blocked in our firewall and we need to seek approval with justification just to use these features"*. This reflects broader concerns about technological readiness, indicating a disconnect between technological advancements and practical implementation, which is also a commonly discussed issue in the literature (Morgan et al., 2022).

Slow adoption of AI tools by some companies is directly attributed to a lack of adequate training, emphasizing the need for more appropriate workforce development initiatives, which is also mentioned by Prasad Agrawal (2023). As one participant emphasized, *"The training of the PM on AI is crucial to project success. The PM who understands how to read reports, enter updates, input testing and end-user training progress will be able to guide kick-off of next steps and go-live events all the better"*. Moreover, despite an emerging interest in generative AI technologies, it does not directly translate into organizational change, signalling a slower progression toward adoption. For example, the University of Iowa kept organizing lighting talks for faculty, students, and staff to enhance to apply of generative AI to their work practice (The University of Iowa Information Technology Services, 2024), but those technology is only adopted to the very limited area, such as automated meeting minutes. One participant highlighted the importance of structured support during this transition, stating, *"It would be helpful to have established consultants and SMEs help with the transition in addition to providing training"*. Finally, confidentiality concerns and organizational policies pose additional challenges and risks. Addressing these issues involves strategic planning, policy development, workforce training and development initiatives, and overcoming operational readiness hurdles, which are crucial for harnessing AI's transformative potential.

5.3. Employee sentiments

Based on the survey responses, employee attitudes toward AI adoption are shaped by a mix of education level, and generational differences. According to the responses, higher-educated individuals often see AI as a means of innovating and enhancing efficiency, while those with less education have a greater fear of job loss. Younger workers tend to be more optimistic about AI's potential to remove monotonous tasks, contrasting with older workers' concerns about job security and data privacy. As one participant observed, *"The younger generation is enthusiastic about the use of AI and sees the benefits it could bring to our processes and procedures if used correctly ... The more seasoned generation is sceptical about the new technology, in terms of job security, ability to provide correct and accurate answers, data security, etc."*. These attitudes are evident in slower-adapting sectors like construction as mentioned in the feedback. One respondent noted, *"The construction industry is generally slower in*

integrating latest technology, so transformation projects sometimes take longer to be implemented. However, it's worth mentioning that some of them do acknowledge the potential benefits it could bring". The reception of AI across workplaces varies, with some fully embracing the technology and others completely rejecting it. Another participant highlighted generational divides within their organization, stating, "We have a mixed age population, so the younglings will easily embrace it, but the rest will take a while". Similarly, another added, "An aging workforce is reluctant to change; our younger staff is far more used to the technology".

Concerns are widespread about data privacy, job security, and ethical issues, alongside scepticism regarding the long-term impact of AI on certain professions. As two individuals noted, "There remains a fear of workforce reductions" and "People are starting to talk about it, but few admit to using it. Fear is a problem" echoing the apprehension that continues to shape workplace attitudes. Despite a few mentioned challenges such as the need for training and infrastructure readiness discussed earlier, the potential benefits such as efficiency improvement and automation are recognized. These diverse sentiments highlight the challenges in achieving a broad acceptance of effective integration of AI in the workplace.

5.4. Improved employee productivity

The theme of productivity in the context of AI integration into the workplace emerges clearly through various participant responses. The employee adoption of AI helps in automating tasks, enhancing data analysis, and enabling employees to concentrate on more complex, value-adding activities. However, an employee learning curve was also acknowledged for adopting AI, which underscores some of the initial challenges. The benefits mentioned include improved communication clarity, especially in written artifacts; reduced manual labour; automated reporting; and more opportunities to focus on critical thinking and continuous improvement for employees. However, generative AI has not yet reached a point where it reliably delivers higher performance (Seo and Kang, 2024), therefore, it requires supervision before being used for tasks involving "hard skills", such as generating contract documents. This balance between automating "hard skills" tasks and human input highlights the complementary nature of AI adoption in the workplace.

5.5. Other thoughts and feedback

The integration of generative AI within project management and sensitive data handling environments as in federal or other governmental agencies present a spectrum of optimism, caution, and acknowledgment of inevitable upcoming change. There is a recognition of AI's potential to significantly save time and expenditure, even as concerns about data sensitivity remain paramount. The sentiments range from viewing AI as a supplementary input that requires human validation, to fear of overreliance and potential devaluation of human roles in project teams. Several respondents highlighted a critical need for maintaining skilled personnel who can critically assess AI-generated content, as it might fail to understand specific nuances of the project. There is an eagerness to engage with AI technologies, underscored by a call for quality use cases and executive-level sponsorship and support to ensure successful adoption. There is a consensus on the transformative impact of AI on project management, with the potential to streamline processes, improve communication, and enhance decision-making.

6. Conclusion

Through workshops and subsequent surveys of 52 project management professionals, the study investigated the impact of these tools on corporate values, employee perceptions, and the applicability of generative AI in various project management processes and roles. The research methodology included a preliminary study, a detailed literature

review, the development of a questionnaire, running workshops to showcase generative AI capabilities, collecting survey data, and the analysis of collected data. The findings highlight the transformative potential of AI in project management, with insights into organizational dynamics, employee sentiments, and the practical applications of AI tools.

Limitations include the relatively small sample size of survey participants, which may not fully represent the diverse range of project management professionals and their experiences with and perceptions of generative AI tools. Recruiting professionals for participation proved challenging due to time constraints and availability, especially among senior-level practitioners. As such, we do not aim for statistical generalizability in the quantitative findings. Rather, the survey results are intended as exploratory and descriptive, offering an early snapshot that complements the qualitative insights from the workshops and surveys. Authors believe that the research framework and mixed-method approach adopted here provide a foundation for future investigations involving larger and more representative samples, which can test the robustness and transferability of the observed patterns. Additionally, the workshops and survey relied on self-reported data, which could be subject to biases and inaccuracies. The study's focus on generative AI tools primarily revolved around ChatGPT, potentially overlooking the nuances and capabilities of other generative AI tools available in the market. The cross-sectional nature of the survey also limits the ability to capture longitudinal changes in perceptions and applications of AI in project management.

The contributions of this work include providing empirical data on the perceptions and applicability of generative AI tools in project management. The study offers practical insights into the transformative potential of AI, addressing both the opportunities and challenges faced by organizations in adopting these technologies. It highlights the critical need for strategic planning, comprehensive training, and addressing ethical, data security, and privacy concerns to leverage AI effectively. Furthermore, the findings underscore the importance of balancing AI-driven automation with human expertise in project management roles, offering practical guidance for future integration efforts.

Future work includes developing quantitative methods for measuring the direct impact of AI on project success rates and employee productivity in a quantifiable manner. Additionally, creating industry-specific frameworks for AI integration in project management will help address the unique challenges faced by different sectors. Further research is also needed to explore the ethical implications of AI adoption, focusing on transparency, bias mitigation, and the development of robust data privacy protocols. The study highlights concern on data security, ethical biases, human-AI interaction, training needs, and implementation challenges, all of which demand further research and development to optimize ChatGPT's role in enhancing project management practices effectively. Addressing these gaps will provide a more comprehensive understanding of AI's potential and its integration into project management practices.

CRediT authorship contribution statement

Vartenie Aramali: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Namho Cho:** Writing – original draft, Resources, Project administration, Investigation, Formal analysis. **Falguni Pande:** Writing – original draft, Project administration, Methodology, Investigation, Conceptualization. **M.K.S. Al-Mhdawi:** Writing – review & editing. **Udechukwu Ojiako:** Writing – review & editing. **Abroon Qazi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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